

What we're doing

Can fold-conditioned Protein generation supply one million useful, structurally diverse, sequence-paired monomer documents of length 60–500, at an acceptable cost per retained structure?

Why

Length-stratified, fold-balanced generation followed by sequence design, refolding and structural selection will increase structural coverage relative to unconditional generation. Whether that coverage improves MarFold must be tested with a fixed-token training comparison; generating more documents alone does not establish value.

Results so far

Issue #278 filed and execution authorized. One-H100 smoke test on cw-rno2a is validating the environment. Five geometry checks pass locally. Four 100-aa backbones generated at 1.803 seconds each (batch 4, uncompiled). Two passed self-consistency and confidence checks and serialized to contacts-v1. ESMFold normalized-confidence units were corrected before filtering. Next: end-to-end smoke, then a pilot capped at 100 H100-hours.